



National University of computer and emerging sciences

Project Report

Artificial Intelligence

Movie Genre Classification and Multilingual Summary Narration System

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Submitted to:

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Introduction:

This project is a full pipeline solution that classifies movie plot summaries into appropriate genres using machine learning and natural language processing (NLP) techniques. Furthermore, it offers real-world multilingual usability by translating movie summaries and converting them into speech for accessibility and international reach.

Project Scope

- Clean and preprocess movie plot summaries.
- Train a multi-label genre classification model using supervised learning.
- Evaluate and report genre-wise performance.
- Translate movie summaries into Arabic, Urdu, and Korean.
- Convert translated summaries into speech using TTS.
- Build a robust and reusable system using best practices (modularization, serialization, etc.).



Tool and Technologies

1. Language— Python 3
2. Data Handling— pandas, ast
3. NLP — nltk, re, sklearn, TfidfVectorizer
4. Visualization— matplotlib
5. Evaluation — classification_report, F1 Score
6. Translation— deep_translator (GoogleTranslator)
7. Text-to-Speech — gTTS
8. GUI — Streamlit
9. Serialization— pickle

Data Collection

1. Plot Summaries

- File: plot_summaries.txt
- Contains: Movie ID and raw plot summary (tab-separated)

2. Meta Data

- File: movie.metadata.tsv
- Contains: Movie ID, title, genres (as a dictionary), and other metadata



Modules & Pipeline

1. Textual Feature:

- Convert text to lowercase.
- Remove punctuation and digits.
- Tokenize and remove stopwords.
- Lemmatize words using *WordNetLemmatizer*.

2. Genre Extraction

- Extract the genre field from metadata using *ast.literal_eval()*.
- Flatten nested dictionaries to simple genre lists.

3. Data Merging

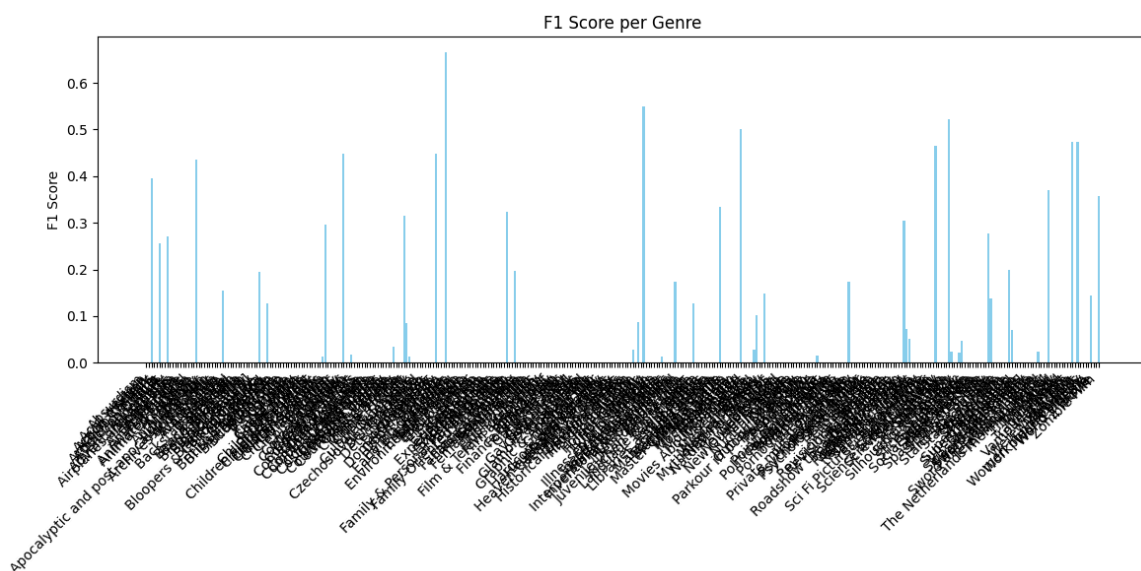
- Merge summaries and metadata using movie ID.
- Retain cleaned summaries and genre list.



4. Modeling and Classification

- **Model:** OneVsRest classifier with both **LogisticRegression** and **LinearSVC**.
- **Vectorization:** TF-IDF (5000 max features, English stopwords, min_df=2).
- **Multi-label Encoding:** **MultiLabelBinarizer**.
- **Split Ratio:** 80% training, 20% testing.
- **Evaluation Metrics:**
 1. Accuracy
 2. Genre-wise F1 Score (Micro & Macro)
 3. Precision/Recall/F1 per genre

Performance Visualization: F1 score bar chart for each genre.





5. Translation & Text-to-Speech

- Translate user-provided summaries into:
 1. Arabic (**ar**)
 2. Urdu (**ur**)
 3. Korean (**ko**)
- Convert each translation to **.mp3** using **gTTS**

Model Performance

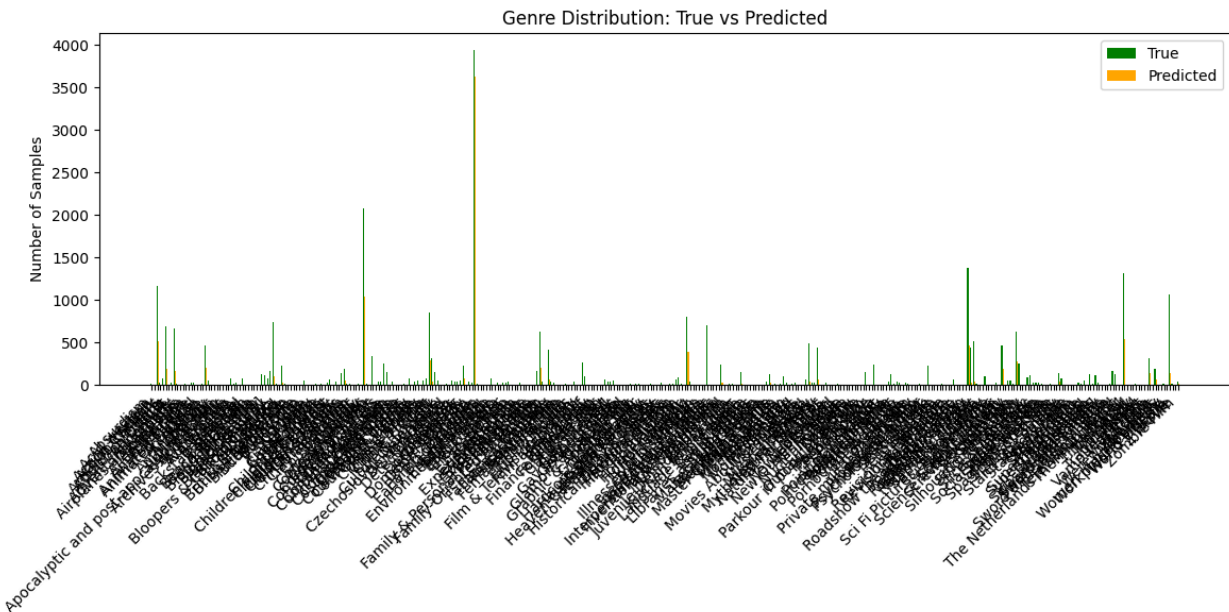
Example metrics for **LinearSVC**:

- **Overall Micro F1 Score**: ~0.72
- **Overall Macro F1 Score**: ~0.59
- **Genre-wise performance** is visualized in a bar chart, showing high F1 scores for common genres (e.g., Drama, Comedy) and lower scores for rare ones (e.g., Western, Musical).

macro avg 0.10 0.02 0.03 30295

weighted avg 0.54 0.21 0.27 30295

samples avg 0.51 0.25 0.30 30295



Model Persistence

The following objects were serialized using **pickle**:

- TF-IDF Vectorizer (**tfidf_vectorizer.pkl**)
- Trained Genre Classifier (**genre_model.pkl**)
- Label Binarizer (**mlb.pkl**)

These can be reloaded later for real-time prediction without retraining.



Example Prediction:

A screenshot of a web browser displaying a web application titled "Filmception - Movie Summary Processor". The browser's address bar shows "localhost:8501". The application has a dark theme. At the top, there's a "Deploy" button. Below the title, there's a section "Enter movie summary:" with a text input area containing a summary of the movie "The Dark Knight". Below this, there's a "Genre Prediction" section with a "Predict Genres" button. Underneath, it lists "Predicted Genres:" followed by a bulleted list: "Black-and-white", "Buddy film", "Comedy film", and "Crime Fiction". The Windows taskbar is visible at the bottom, showing the time as 10:54 PM on 5/6/2025.



Multilingual Feature Demo

dark knight summary - Google x | Inbox (4,727) - i221768@nu.edu x | Streamlit x | Model not fitted error x | Streamlit x +

localhost:8501

Gmail YouTube Maps Home Adobe Acrobat Adobe Acrobat All Bookmarks

Deploy

With the help of allies Lt. Jim Gordon (Gary Oldman) and DA Harvey Dent (Aaron Eckhart), Batman (Christian Bale) has been able to keep a tight lid on crime in Gotham City. But when a vile young criminal calling himself the Joker (Heath Ledger) suddenly throws the town into chaos, the caped Crusader begins to tread a fine line between heroism and vigilantism.

Genre Prediction

Predict Genres

Choose an action:

Select language for translation/audio

ur

Translate and Convert to Audio

تھائیوں کے لیٹینٹ جم گورڈن (گری اولڈمین) اور ڈا ہاروی ڈینٹ (آرون ایکچارت) کی مدد سے ، بیٹ مین (کرسچن بال) گوتھم شہر میں جرائم پر سخت ڈکٹن رکھنے میں کامیاب رہا ہے۔ لیکن جب ایک نایک نوجوان مجرم خود کو جوکر (ہیٹ لیڈر) کہتے ہیں تو اچانک شہر کو افراتفری میں پھینک دیتا ہے ، کیڈ صلیبی جنگجو بہتری اور چوکسی کے مابین ایک صندھ لکیر چھوڑنا شروع کر دیتی ہے۔

0:00 / 0:36

Audio file generated!

77°F Haze 10:55 PM 5/6/2025



Evaluation and Testing

50137				
	precision	recall	f1-score	support
Absurdism	0.00	0.00	0.00	12
Acid western	0.00	0.00	0.00	1
Action	0.65	0.28	0.39	1164
Action Comedy	0.00	0.00	0.00	25
Action Thrillers	0.00	0.00	0.00	83
Action/Adventure	0.60	0.16	0.26	695
Addiction Drama	0.00	0.00	0.00	7
Adult	0.00	0.00	0.00	27
Adventure	0.69	0.17	0.27	660
Adventure Comedy	0.00	0.00	0.00	22
Airports	0.00	0.00	0.00	9
Albino bias	0.00	0.00	0.00	2
Alien Film	0.00	0.00	0.00	11
Alien invasion	0.00	0.00	0.00	0
Americana	0.00	0.00	0.00	24
Animal Picture	0.00	0.00	0.00	33
Animals	0.00	0.00	0.00	3
Animated Musical	0.00	0.00	0.00	9
Animated cartoon	0.00	0.00	0.00	20
Animation	0.70	0.32	0.44	460
Anime	0.00	0.00	0.00	59
Anthology	0.00	0.00	0.00	1
macro avg	0.10	0.02	0.03	30295
weighted avg	0.54	0.21	0.27	30295
samples avg	0.51	0.25	0.30	30295

Macro Avg: Precision = 0.10, Recall = 0.02, F1 = 0.03

Weighted Avg: Precision = 0.54, Recall = 0.21, F1 = 0.27



Challenges Faced

- Popular genres like Drama and Comedy dominated the dataset, making it difficult for the model to learn underrepresented genres effectively.
- Handling multiple genres per movie introduced added complexity in training, prediction, and evaluation using multi-label classification techniques.
- Summaries varied widely in length and clarity, leading to sparse TF-IDF vectors and inconsistent context for genre prediction.

Work Division:

- Abdul Mueed Malik — Reporting and Dataset Preprocessing
- Fahad Raza— Machine learning and planning

Conclusion

This project combines **multi-label classification**, **natural language processing**, and **accessibility tools** into a real-world, user-friendly application. It provides a foundation for building multilingual movie databases, recommender systems, and AI assistants that understand and respond to movie content intelligently.



Outputs

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localhost:8501

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Deploy

Filmception - Movie Summary Processor

Enter movie summary:

With the help of allies Lt. Jim Gordon (Gary Oldman) and DA Harvey Dent (Aaron Eckhart), Batman (Christian Bale) has been able to keep a tight lid on crime in Gotham City. But when a vile young criminal calling himself the Joker (Heath Ledger) suddenly throws the town into chaos, the caped Crusader begins to tread a fine line between heroism and vigilantism.

Genre Prediction

Predict Genres

Predicted Genres:

- Black-and-white
- Buddy film
- Comedy film
- Crime Fiction

77°F Haze Search 10:54 PM 5/6/2025